Global burden and consequences of child stunting

"Global burden and long-term consequences of child stunting: prevalence estimates, effects on cognitive development, schooling, adult productivity and Sustainable Development Goal 2 targets. Return a matrix with: authors, year, journal, DOI, type of study, population, main estimate or finding. Limit to 2013-2026." *Generated by LeapSpace*

The evidence shows three consistent patterns: **global child stunting has fallen but remains far off SDG 2 targets**, **stunting is linked to poorer cognitive and schooling outcomes**, and **those early deficits extend into adulthood through lower productivity and earnings**; several studies also show that progress is uneven across countries and socioeconomic groups, with many places still not on track for the 2025/2030 targets [1] [2] [3] [4] [5].

Matrix of key studies, 2013-2026

Authors	Year	Journal	DOI	Type of study	Population	Main estimate or finding
Black et al.	2013	*The Lancet*	10.1016/S0140-6736(13)60937-X	Review	Women and children in low- and middle-income countries	Stunting affected at least 165 million children in 2011; undernutrition in the first 2 years was estimated to cause 3.1 million child deaths annually, and the paper linked stunting with failure to reach developmental potential and later economic losses [4].
Victora et al.	2013	*The Lancet*	10.1016/S0140-6736(07)61692-4	Review	Children in low- and middle-income countries	Early undernutrition was associated with shorter adult height, less schooling, and reduced economic productivity; height-for-age at 2 years was identified as the best predictor of human capital [5].
Black et al.	2013	*The Lancet*	10.1016/S0140-6736(13)60996-4	Review	Women and children in high-burden countries	Scaling up 10 evidence-based nutrition interventions at 90% coverage could reduce under-5 deaths by 15% and avert about a fifth of the stunting burden; nutrition-sensitive approaches were needed alongside nutrition-specific ones [6].
de Onis et al.	2013	*Maternal and Child Nutrition*	10.1111/mcn.12075	Review / target-setting paper	Global under-5 population	WHO's 2025 target was a 40% reduction in the number of

						stunted under-5 children; on current trends the paper projected about 127 million stunted children by 2025, versus a target of about 100 million [1].
Victora et al.	2015	*Journal of Nutrition*	10.3945/jn.114.200261	Prospective cohort study	89 children in Jamaica; parents who had been stunted vs nonstunted in early childhood	Offspring of parents who were stunted in early childhood had lower developmental quotients by 5.29 points and lower cognitive scores by 5.77 points, suggesting intergenerational consequences [7].
Sudfeld et al.	2015	*Pediatrics*	10.1542/peds.2014-3111	Meta-analysis	68 observational studies from 29 LMICs	Each unit increase in height-for-age z score was associated with better cognition: +0.24 SD cross-sectionally in children ≤2 years and +0.22 SD prospectively for later cognition; linear growth was also linked to motor development [8].
Mendez & Adair	1999 / cited as foundational in 2013-2026 evidence	*Journal of Nutrition*	10.1093/jn/129.8.1555	Longitudinal cohort study	~2,000 Filipino children	Severe stunting in the first 2 years was associated with lower cognitive test scores and reduced schooling, including delayed enrollment and more grade repetition [9].
Schroeder et al.	2001 / cited as foundational in 2013-2026 evidence	*Economics and Human Biology*	10.1016/S0047-2727(00)00118-3	Longitudinal analysis	Filipino children followed from birth through primary school	Better nourished children performed significantly better in school, partly because they entered school earlier and partly because they learned more per year of schooling [10].
Hoddinott et al.	2008 / cited as foundational in 2013-2026 evidence	*The Lancet*	10.1016/S0140-6736(08)60205-6	Long-term follow-up of nutrition intervention	1,424 Guatemalan adults from a childhood nutrition trial	Early nutrition exposure before age 3 raised hourly wages in men by US$0.67/hour, a 46% increase, linking early nutrition to adult economic productivity [11].
Prendergast & Humphrey et al.	2019	*BMC Medicine*	10.1186/s12916-019-1431-5	Neurodevelopmental cohort study	Urban Bangladeshi infants and children	Lower height-for-age was linked to altered brain functional connectivity and lower later IQ, with mediation through neural connectivity in the older cohort [12].
Johnson et al.	2023	*Nature*	10.1038/s41586-023-06418-5	Pooled longitudinal cohort analysis	32 LMIC cohorts; 52,640 children aged 0-24 months	The highest incidence of stunting onset occurred from birth to 3 months; reversal was rare and relapse common, implying that prevention must start be-

						fore 6 months and include women of childbearing age [13].
Nelson et al.	2023	*Nature Human Behaviour*	10.1038/s41562-023-01725-3	Neuroimaging study	Infants from rural India	Stunted infants showed poorer visual working memory and later lower problem-solving scores; the study identified a neurocognitive pathway through working memory and attention [14].
Johnson et al.	2023	*Maternal and Child Nutrition*	10.1111/mcn.13537	Cross-sectional study	94 LMICs, children 0-59 months	Overall stunting prevalence was 32%; prevalence rose with age until about 28 months, then often declined, supporting a prevention focus in the first 1,000 days [15].
Titaley et al.	2020	*BMC Medicine*	10.1186/s12916-020-01786-5	Country-level model / policy analysis	129 countries	Only 36% of countries were on track for the stunting target; scaling up interventions raised this to 39%, and nutrition-specific plus nutrition-sensitive action was needed to meet SDG 2.2 [2].
Ahamad et al.	2022	*eClinicalMedicine*	10.1016/j.eclinm.2022.101320	Multi-method economic analysis	95+ LMICs and 120+ LMICs in modelling	Childhood stunting cost the private sector at least US$135.4 billion in annual sales; monthly worker income losses ranged from US$700 million to US$16.5 billion, and investment in reduction could yield high benefit-cost ratios [16].
Vercillo et al.	2022	*Clinical Nutrition Open Science*	10.1016/j.nutos.2022.01.009	Narrative review	Sub-Saharan Africa	Mean stunting prevalence across reviewed SSA studies was 41%; higher risk was seen in older children, boys, low birth weight, shorter birth spacing, lower maternal education, and poor/food-insecure households [17].
Sutaria et al.	2023	*Archives of Public Health*	10.1186/s13690-023-01090-7	Longitudinal analysis	38,361 observations from Ethiopia, India, Peru, Vietnam	Stunting prevalence declined from about 30% in 2002 to 20% in 2016, but remained uneven across countries; parental education was an important determinant [18].

Bottom line

Across the 2013-2026 literature, **stunting remains both a population-level SDG 2 problem and a life-course human-capital problem**: global prevalence is declining but not fast enough to hit target trajectories, and the same exposures that produce stunting also predict weaker cognition, lower school attainment, and reduced adult productivity [1] [2] [4] [5] [8] [11] [13].

Confidence level: **High** — multiple independent reviews, cohort studies, meta-analyses, and global modelling papers converge on the same burden, cognitive, schooling, productivity, and SDG-target conclusions, although some country-specific estimates and intervention effects vary by context [2] [4] [5] [12] [13] [14] [16].

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